

Xinyi Yang

Robotics Hardware | Mechanical Design | Prototyping & Testing

Atlanta, GA | xinyiyangcindy000@gmail.com | [linkedin.com/in/yang-xinyi-intro](https://www.linkedin.com/in/yang-xinyi-intro) |

xinyiyang0.github.io/engineering-portfolio/

U.S. Permanent Resident | No sponsorship required

SUMMARY

Robotics PhD researcher at Georgia Tech with experience developing robotic and adaptive mechanical systems from concept and CAD design through fabrication, integration, and experimental validation. Skilled in rapid prototyping, mechanical testing, structural dynamics, and iterative design. Seeking a Spring or Summer 2027 internship in robotics hardware, mechanical design, or mechanical systems engineering.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Aerospace Engineering, Robotics

Expected 2028

Research Focus: Robotics hardware, adaptive mechanical systems, and structural dynamics

Harvard University

Cambridge, MA

Master in Design Studies

2023

Focus: Digital Fabrication, Material Systems, and Robotics

Cross-registration: Massachusetts Institute of Technology, Mechanical Engineering

South China Agricultural University

Guangzhou, China

Bachelor of Engineering in Architecture

2017

Focus: Structural Design, Materials, and Digital Fabrication

SKILLS

Mechanical Design & CAD: SolidWorks, Fusion 360, Rhino, AutoCAD

Manufacturing & Prototyping: CNC milling, FDM and SLA 3D printing, laser cutting, silicone mold making and casting, vacuum forming, injection molding, composite lay-up, fixture fabrication

Testing & Experimental Analysis: Instron mechanical testing, pressure-chamber testing, vibration testing, wind-tunnel testing, high-speed imaging

Programming & Data Processing: Python, MATLAB, ROS 2, Arduino, Origin, RStudio

RESEARCH EXPERIENCE

Computational Design and Metamaterial Lab, Georgia Institute of Technology

Atlanta, GA

Graduate Researcher

Mar 2024 – Present

- Designed frequency-addressed modular connectors for electronics-free robotic structures, tuning geometry and stiffness to enable selective vibration-triggered disassembly and sequential deployment.
- Characterized connector strength, resonant response, and release thresholds through mechanical loading and vibration testing, translating experimental results into design rules for multi-module prototypes.
- Developed a printer-integrated, resonance-based testing workflow to characterize FDM-printed materials, reconstructing elastic modulus, plastic response, fracture strain, and damping from high-speed imaging and vibration data.
- Validated reconstructed material properties against conventional mechanical testing across PLA, PETG, recycled, and composite filaments, achieving less than 3% modulus difference for geometry-matched PLA and PETG specimens.
- Designed, fabricated, and wind-tunnel tested bistable, seed-inspired mechanisms, quantifying deployment thresholds and aerodynamic response across multiple geometries and mounting orientations.
- Established fabrication and mechanical-testing capabilities for a new robotics laboratory, including equipment specification, purchasing, laboratory layout, calibration, and experimental workflow development.

Morphing Matter Lab, Carnegie Mellon University

Pittsburgh, PA

Research Associate

Sep 2023 – Dec 2023

- Redesigned and fabricated second-generation bio-inspired seed carriers using sustainable composite materials, doubling mechanical lifespan through material selection, structural redesign, and additive-manufacturing process optimization.
- Developed a scalable fabrication and assembly workflow that increased production from approximately one unit per week to hundreds of reproducible units, enabling independent production by undergraduate teams.

Micro-Robotics Laboratory, Harvard University

Cambridge, MA

Research Assistant

May 2022 – Jun 2023

- Fabricated and assembled a suction-based marine adhesion system for autonomous sensor deployment, integrating vacuum-generation, sealing, and burn-wire release subsystems into functional hardware prototypes.
- Conducted pressure-chamber testing up to 900 psi and long-duration submersion trials to evaluate sealing integrity, structural durability, and release reliability under simulated deep-sea conditions.

Computer Science and Artificial Intelligence Laboratory, MIT

Cambridge, MA

Research Assistant

May 2022 – Jun 2023

- Designed and fabricated laminated composite sensor-actuator sheets integrating structural substrates, programmable magnetic layers, and flexible PCBs for shape-changing interactive hardware.
- Developed laser-cut fabrication, alignment, lamination, and assembly workflows to support rapid iteration and repeatable integration of embedded electronic and mechanical components.
- Built UWB- and ESP32-based wireless localization and communication prototypes for VR and IoT applications, evaluating angular resolution and iterating anchor and tag geometry for manufacturability and deployment.

PROFESSIONAL EXPERIENCE

LWK + Partners

Hong Kong

Project Manager

Oct 2017 – Dec 2020

- Coordinated architectural, structural, mechanical, electrical, and plumbing systems across 15+ stakeholder teams for large-scale healthcare and government infrastructure projects.
- Managed design development, technical documentation, multidisciplinary coordination, and construction-stage implementation under regulatory, fabrication, safety, and sustainability constraints.
- Resolved cross-disciplinary design conflicts with consultants, contractors, and clients, improving constructability and reducing documentation revisions and on-site rework.

PUBLICATIONS

- Yang, X., et al. "Remote Disassembly of Electronics-Free Modular Structures." *Nature Communications*, 2026.
- Yang, X., et al. "Electronic-Free Particle Robots Communicate through Architected Tentacles." *Advanced Intelligent Systems*, 2025.
- Chen, F., Yang, X., et al. "Subhertz Band Gaps in Ultralightweight Inerter-Based Metamaterials." *Physical Review Applied*, 25(2), 024032, 2026.
- Meyer, G., Vogt, D. M., Yang, X., et al. "Passive Vacuum Regeneration for Suction-Based Marine Adhesion Devices." *IEEE Journal of Oceanic Engineering*, 2025.